A Project Report

on

Extractive Text Summarization of News Articles

Bachelor of Technology

IN

COMPUTER ENGINEERING

BY

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Dated: 19/06/2020

Declaration

This is to certify that the work presented in project entitle "Extractive Text Summarization of News Articles" submitted to the Department of Computer Engineering, Zakir Husain College of Engineering and Technology, Aligarh Muslim University Aligarh, for the award of the degree of Bachelor of Technology in Computer Engineering, during the session 2019-20, is our original work. We have neither plagiarized nor submitted the same work for the award of any degree.

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Certificate

This is to certify that the Project Report entitled "Extractive Text Summarization of News Articles", being submitted by "Areeba Aziz and Lakshita Mukul", in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Engineering, during the session 2019-20, in the Department of Computer Engineering, Zakir Husain College of Engineering and Technology, Aligarh Muslim University Aligarh, is a record of candidates' own work carried out by them under our supervision and guidance.

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Table of Contents

Abstract

The World Wide Web (WWW), commonly known as the Web, is an information system where documents and other web resources are identified by Uniform Resource Locators (URLs), and are accessible over the Internet. Information size on the WWW and in other electronic form is increasing tremendously. This includes business transactions, news reports, satellite data, digital media, text reports and memos and biological information. News is information about current events happening all around the world. It plays an important role as it adds up to people's worldview and helps them stay connected in the society and knowing when and how to form opinions and judgment. There was a time when people used to wait for newspapers in order to catch the previous day's happenings but thanks to the internet, all the latest information is now available with a click of a button. However, the information is much larger than one can manage, easily and efficiently, also today everyone wants to gain more information in less time. Instead of reading large documents and then getting the insight, it is better to read summary which gives the core information about the topic and helps in gaining more information in less time. Therefore there is a need for some form of information compression which can be achieved by various mining tasks like classification, clustering and summarization that help in understanding the information. Summarization is capturing the essence of the document in fewer sentences without affecting the structure of the original document. Summarization is of two types- manual and automatic. Manual is written by human experts, which is time consuming and tedious. Automatic summary is when the summary is automatically generated by the system using some algorithm. Automatic summary is of two types- extractive and abstractive. In extractive summary the most important sentences which are selected on the basis of the technique being used, forms the summary while in abstractive summary the summary is created without copying the sentences from the original document by preserving its structure. Connectivity among the sentences is major issue because of which most of the users don't rely on system generated summary. In our project we have implemented three techniques for generating the extractive summary of news articles of the two benchmark datasets- CNN-corpus and the BBC (these datasets have both the article and its summary) on different retention rates in order to know what would be the best retention rate for generating the summary which would contain most of the information of the original text without much affecting the connectivity among the sentences since, readability and connectivity are the two prime factors because of which most of the people still rely on man written summaries. For this we have performed both qualitative and quantitative evaluation on the generated summaries. The quantitative evaluation was done using ROUGE. This evaluation was done in order to know how much of the generated summary matches the summary given in the dataset while in qualitative evaluation we checked the readability and connectivity of the generated summary. We have also compared the results of the evaluation with that of the base paper we referred. After the evaluation we critically analyzed the results and gave several observations about the relation of the summary generated with a particular retention rate with that of the parameters such as precision and recall which affects the overall quality of the summary. In the end we have given the retention rate that generated the best summary among all in terms of being able to give gist of the original article while maintaining the connectivity among the sentences. We have also proposed several methods which if incorporated in extractive text summarization system will capture the most relevant information regarding the article given that the article is a News article since in our project we have dealt with News articles only.

Acknowledgements

First of all we would like to thank the Almighty, who has always guided us to work on the right path of the life. Our greatest thanks are to our parents who bestowed ability and strength in us to complete this work. We wish to express our deep gratitude to our mentors Ms. Hira Javed and Mr. Tameem Ahmad for their valuable guidance and suggestions throughout the process of accomplishing our report and giving helpful guidelines about our project. We must pay acknowledge to our various teachers who taught us various subjects of "Computer Engineering" throughout the batch.

Lakshita Mukul

Areeba Aziz

Date: 19 June 2020

Acronyms

TF-IDF	Term Frequency Inverse Document Frequency
TF	Term Frequency
IDF	Inverse Document Frequency
SVM	Support Vector Machine
BBC	British Broadcasting Corporation
CNN	Cable News Network
ROUGE	Recall-Oriented Understudy for Gisting Evaluation
LCS	Longest Common Subsequence
NN	Noun, singular or mass
NNS	Noun, plural
NNP	Proper noun, singular
NNPS	Proper noun, plural
GloVe	Global Vectors
MB	Mega Byte
NLTK	Natural Language Toolkit
csv	Comma-Separated Values
os	Operating System
re	Regular Expressions

List of Figures

List of Tables

Chapter 1 : Introduction

This chapter briefly introduces the project work proposed in this report. Section 1.1 gives an overview of the research undertaken. Section 1.2 discusses automatic text summarization and then various methods of summarization are briefly described in Section 1.3. Section 1.4 explains the motivation behind the proposed method, the objective and the scope of the method is enlighten in Section 1.5. Section 1.6 presents an outline of this report and labeling the remaining chapters. Finally, Section 1.7 gives the summary of the chapter.

1.1 Overview

According to www.worldwidewebsize.com, the indexed Web contains at least 5.47 billion pages (Sunday, 18 June, 2020). With the massive proliferation in the velocity, volume and variety of information accessible online and the consequent need to develop viable paradigms which facilitate better techniques to access this information, there has been a strong resurgence of interest in Web Information Retrieval (Web IR) research in recent years. With the ease in availability of Internet and increased use of smart devices like mobile phone, laptop, tablets the access and storage of data has escalated rapidly, resulting in need of tools which can enhance the user's productivity and experience.

Text summarization is the process of reducing the huge amount of data which is available on Internet or on any other source of information like newspapers, books etc., into summarized documents, in order to create the summary of the document that retains the most important points and the overall meaning of the document. Summarization has been identified as an effective method, which helps users to locate the right information at the right time thus facilitating timely decisions.

Automatic text summarization techniques can be used for extracting keyword. Extracting keywords using text summarization algorithm can optimize the summarization process. However, several text summarization methods have their own pros and cons. We have used three techniques namely TextRank, Lexical Chain and TF-IDF in our project for survey. The details of the techniques have been discussed in Chapter 2.

1.2 Automatic Text Summarization

Automatic Text Summarization is the process of finding a summary of a document available on the Web by a computer without changing the meaning of the original document for retrieving the useful information like the structure of document[11]. An automated generated summary can be of various types depending on the purpose and kind of data available for summarization. This categorization of text summarization techniques is depicted in the Figure 1.1 below[9].

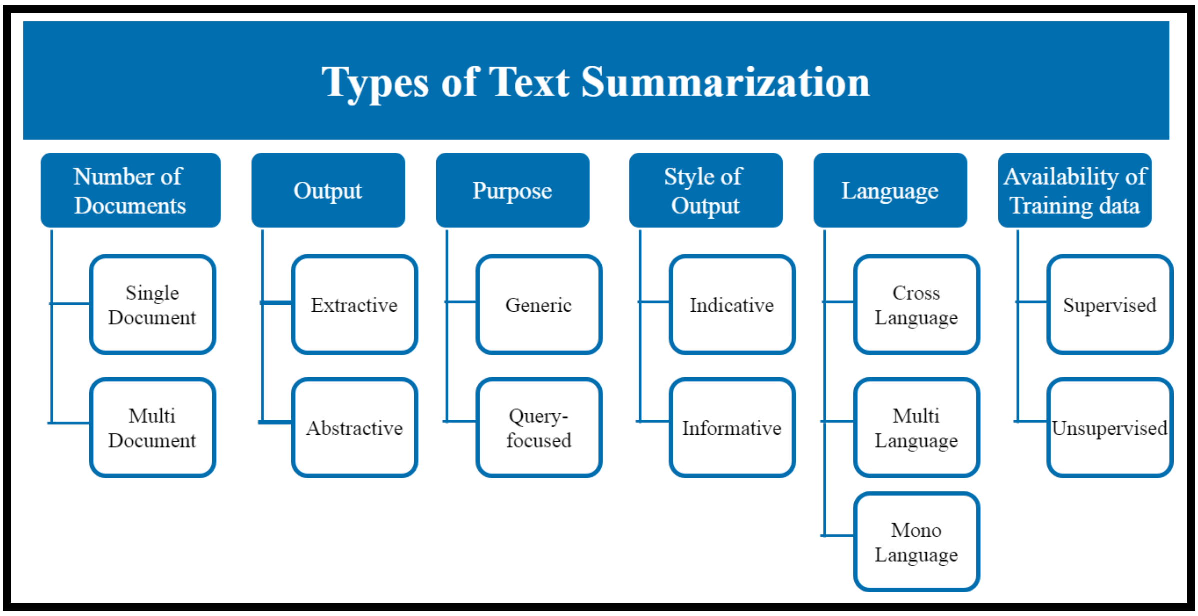

Figure 1.1 : Text Summarization Categories based on different characteristics

Primary categorization of text summarization techniques is on the basis of the type of summary generated. It can either be of abstractive or extractive type. In Abstractive Technique, the main concept of the document is understood then those concepts are expressed in precise form using simple natural language. In this technique the generated summary doesn't include same sentences from input text as compared to the extractive technique.

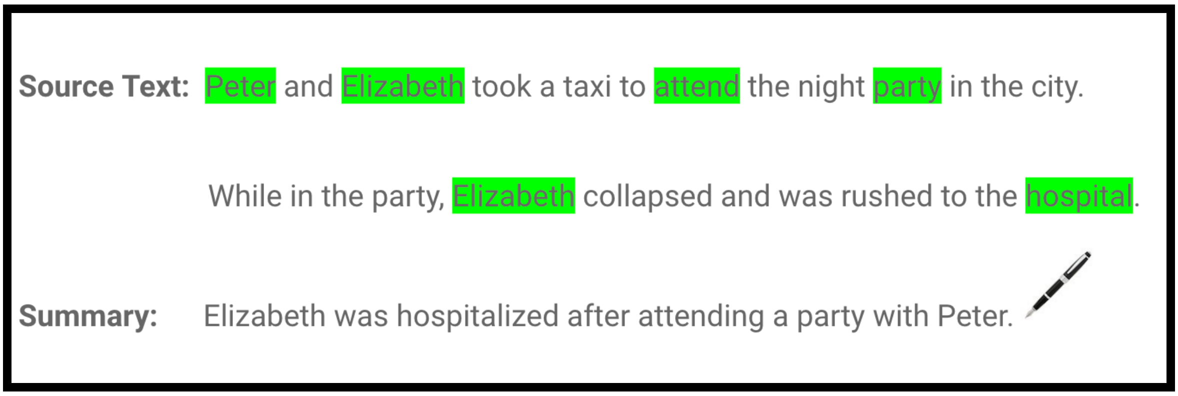

Figure 1.2 : Example of Abstractive Text Summarization

On the other hand, Extractive text summarization uses most important sentences from the document exactly as they appear in it and then integrated in shorter form. The importance of sentences is decided on the basis of statistical and linguistic features of sentences. Most of the studies on text summarization are on extractive techniques[10].

My husband (who, being Mexican, is very picky about his tortilla chips) and I absolutely love these! The texture is light and crispy, rather than thick and crunchy. He actually usually prefers a very hearty, crunchy chip (Like El Ranchero), but the flavor of these is so fantastic that we're both thrilled with them. The bean, rice and corn base makes them incredibly flavorful, and they have a touch of onion and garlic in addition to that. We go through an embarrassing amount of them. I never, ever like plain chips, but these I can eat without anything else, although they're particularly amazing with a fresh salsa. I highly recommend these!

SUMMARY:

The bean, rice and corn base makes them incredibly flavorful, and they have a touch of onion and garlic in addition to that. I never, ever like plain chips, but these I can eat without anything else, although they're particularly amazing with a fresh salsa. I highly recommend these!

Figure 1.3 : Example of Extractive Text Summarization

1.3 Summarization Methods

Due to the inherent complexity of generating abstractive summary, extractive summaries have been more frequently generated and used in practical applications[9]. Various types of methods can be used to generate extractive summaries such as statistical based methods, graph-based methods, discourse based methods, topic based methods, machine learning based methods and swarm based or optimization based techniques[10].

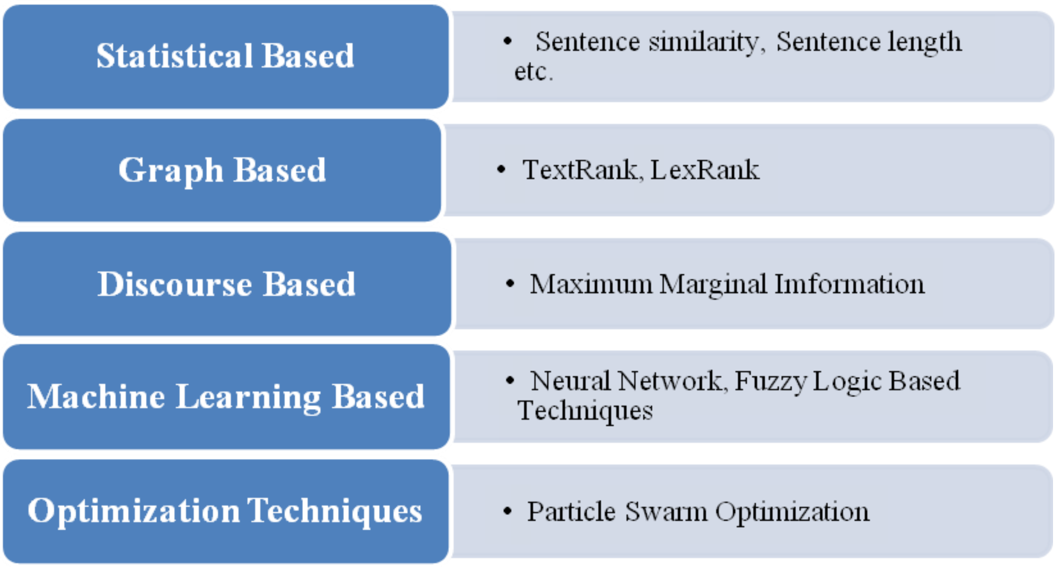

Figure 1.4 : Summarization Methods

These methods have been briefly explained below.

- **Statistical Based Methods:** The methods generate summaries using statistical features of the document like sentence position, centrality of the sentence, sentence length, numeric data in sentence, title similarity etc.[11]. These techniques do not require much storage or fast processors and that language independent.

- **Graph Based Methods:** In this method, the words or sentences are represented by nodes of the graph and edges between these node represents the similarity value between these nodes. The sentences to be taken in extractive summary are found by traversing the graph and selecting the sentences which have similarity index above the defined threshold. Some of the recognized graph based methods are TextRank and LexRank[8].
- **Discourse Based Methods:** These methods require understanding the textual structure and are complex to use as they take into account the connections between sentences and parts in a text[11].
- **Topic Based Methods:** In this method the summary is generated by firstly identifying the subject or theme of the document. Then this is used to extract the sentences which are related to the subject[11].
- **Machine Learning Based Methods:** These include approaches where the machine learn to produce the summary from data provided to it. These data can either be supervised where the training data is provided with the summary and the machine can learn how to produce the summary from this training data or unsupervised where the machine learns by analyzing the document since no training data is available. Some of the machine learning techniques are neural network, SVM, Genetic algorithm and fuzzy logic[8].
- **Optimization techniques:** These techniques use nature inspired algorithms such as swarm algorithms and are usually used in combination with other techniques[10].

1.4 Motivation

Due to fast growing of Internet, there is high availability of the information regarding current affairs but it is very difficult for a human to find the relevant information from the huge pile of both relevant and irrelevant information regarding a particular news. Also in this fast growing era, we do not have enough time to read a large article for a single topic. Therefore, there is a need of text summarization so as to get the gist of the news in minimum words possible in order to save the time and the human efforts. The need of an hour is that we need to built an efficient system which automatically retrieves the document from the huge pile of information, categorize it according to the information need of the user and summarize the document as per user's requirements.

1.5 Objective and Scope

With the help of text summarization process less effort and less time of the human being is used since a person get the gist of the document just by reading the summary of the same. Text summarization is of two types which are- Manual summarization and Automatic summarization. Manual summarization is the process through which the summary of the large document is created manually i.e. with the help of human experts, which is a very difficult task. Manual summarizations have mainly two disadvantages, firstly, the relevant document has to be searched from the Internet where large number of documents on the same topic is available and secondly this whole procedure of manual summarization is time consuming and may be stressful for the human expert. Due to these disadvantages of Manual Text Summarization, Automatic Text Summarization came into being and in the present scenario only automatic text summarization is used. Automatic summarization is the process of creating a summary of large document available on the Internet automatically with the help of a computer program without affecting the overall meaning of the original document.

However, the efficiency of given system widely depend upon several factors such as the algorithm being used, the datasets, the retention rate, etc. In our project we have implemented three different text summarization techniques namely Lexical Chains, TF-IDF and TextRank and performed comparative analysis of three algorithm on two different datasets of news articles. We have also compared the result obtained from various these automatic extractive text summarization techniques with that of survey paper[2].

1.6 Organization

The main body of the report is preceded by detailed contents including lists of figures, lists of tables, and acronyms used. This is followed by the executive summary giving briefly the scope and objectives of the project, importance of the topic, methodologies and major observations.

Chapter 1 presents the overview, scope and motivation of the project.

Chapter 2 provides the insight to some of the research work that has already been done and has helped us to understand the topic deeply and also about the concepts used in this project.

Chapter 3 gives the details of the methodology employed.

Chapter 4 describes the implementation of algorithms. It discusses all the input sets, platform and tool used to implement the technique, to produce result and to compare them.

Chapter 5 describes the experimental results obtained from the input sets used. It presents the analysis of the tests performed.

Chapter 6 presents conclusion and future scope based on the contributions made by this report.

1.7 Chapter Summary

This chapter presents the idea used in this report. It discusses problem, objectives, goals and motivation for our endeavor. Justification for the problem is outlined, together with an explanation of the methodology used in our endeavor. The next chapter describes the literature survey and relevant background study done by us in context of our project.

Chapter 2 : Literature Review

This chapter describes the background work that we have gone through before undertaken this project. Section 2.1 describes the paper that we have studied. Section 2.2 describes the Automatic Text Summarization using Lexical Chain. Section 2.3 describes the Automatic Text Summarization using TF-IDF. Section 2.4 describes the Automatic Text Summarization using TextRank. Finally, Section 2.5 gives the summary of the chapter.

2.1 Papers Referred For Our Endeavor

Paper 1 : Automatic Text Summarization of News Articles[1].

In this paper, the author proposed a technique using a model of topic progression derived from lexical chains to generate coherent summaries by identifying the most important portions of the document. The author optimized the algorithm by implementing pronoun resolution and enhanced sentence scoring to leverage the structure of news articles.

The author also proposed improvement to the existing system by adding adjectives in the lexical chains along with nouns and then observe the effect on the generated summary as a future work.

Paper 2 : Assessing sentence scoring techniques for extractive text summarization [2].

In this paper, the author performs a qualitative and quantitative assessment of 15 algorithms for sentence scoring which is the technique most used for extractive summarization. The evaluation was done on three different datasets News, Blogs & Article contexts. The evaluation was done using ROUGE which is used to quantitatively evaluate the generated summaries.

The author also proposed several strategies such as Stemming, Truncation, Lemmatization, Removal of stop words etc. using to improve the sentence scoring techniques.

2.2 Automatic Text Summarization Using Lexical Chain

A lexical chain is a logical group of semantically related words which depict an idea in the document. The relation between the words can be in terms of synonyms, identities and hypernyms/hyponyms[1]. For example, we can put words together when:

- Two noun instances are identical, and are used in the same sense. (*The cat in the kitchen is large. The cat likes milk.*)
- Two noun instances need not be identical but are used in the same sense (i.e., are synonyms). (*The bike is red. My motorcycle is blue.)*

- The senses of two noun instances have a hypernym/hyponym relation between them. A hypernym is a word with a broad meaning constituting a category into which words with more specific meanings fall. (*Daniel gave me a flower. It is a Rose.*)

- The senses of two noun instances are siblings in the hypernym/hyponym tree. (*I like the fragrance of Rose. However Sunflower is much better.*)[1]

2.3 Automatic Text Summarization Using TF-IDF

TF-IDF is intended to reflect how important a term is to a given document in a collection. It is a numeric measure that is use to score the importance of a word in a document based on how often did it appear in that document and a given collection of documents. The intuition behind this measure is : If a word appears frequently in a document, then it should be important and we should give that word a high score. But if a word appears in too many other documents, it's probably not a unique identifier, therefore we should assign a lower score to that word.

Formula for calculating TF and IDF:

TF(w) = (Number of times term w appears in a document) / (Total number of terms in the document)

IDF(w) = log_{10}(Total number of documents / Number of documents with term w in it)

Hence TF-IDF for a word can be calculated as:

*TF-IDF(w) = TF(w) * IDF(w)*

2.4 Automatic Text Summarization Using TextRank

It belongs to the category of extractive summarization techniques which constructs the summary by extracting the most important sentences from the original text. TextRank algorithm is a simple application of PageRank algorithm. The graph is built using natural language processing which establishes the relationship between the entities of the text. A vertex gains more importance if it forms higher number of links with other vertices because when one vertex links with another, it casts a vote in favor of that vertex. Thus the score of the vertex is calculated by considering the inbounds and outbounds of that vertex. The graph based ranking of the vertices in the graph can be determined by evaluating the associated score of each vertex[12].

The score of a vertex Vi is defined by the following equation:

$$S(Vi) = (1-d) + d * \sum_{j \in In(Vi)} \frac{1}{|Out(Vj)|} S(Vj)$$

Where, G = (V , E) represents a directed graph with the set of vertices V and set of edges E. In (Vi) and Out (Vi) represents the in-bounds and out-bounds for a vertex Vi and d $\in$ [0, 1] is a damping factor in which d is a probability the node visits neighboring node and (1 − d) is the probability of jumping from one vertex to some random vertex. It is generally assigned as 0.85 [12].

However, we can’t measure the strength of the connection between two vertices. Thus, a new formula is introduced which incorporates the weight of the edges while computing the score of vertex and is given as:

$$WS(Vi) = (1 - d) + d * \frac{wij}{\sum_{Vk \in Out(Vj)} wjk} * WS(Vj)$$

where, $WS\ (Vi)$ - weighted score for vertex Vi.
wij - weight which represents the strength of the connection between two vertices Vi and Vj.

2.5 Chapter Summary

This chapter discussed about the background study that we have done and basic concepts and knowledge required to build our project.

Chapter 3 : Proposed Work

This chapter illustrates the approaches that we used in our project. Section 3.1 gives the overview of the research undertaken. Section 3.2 portraits the architectural view of our work. Section 3.3 describes each module of the system and how all the algorithm works. Lastly, Section 3.4 gives the summary of the chapter.

3.1 Proposed Framework

In our project, we have implemented three different algorithms i.e., Lexical Chains, TF-IDF and TextRank and have generated the summaries on different retention rate i.e., 30-percent, 40-percent and 50-percent on two different benchmark datasets namely BBC news dataset and CNN news dataset. All the three algorithm has been run on all the articles present in both datasets for three different retention rate.

From each dataset, we picked up the articles one by one and has performed some pre-processing on it. After preprocessing, we extract the keywords which gives the central idea of the article and based on these keywords, we score the individual sentence and then we have selected the top n sentences to form the summary. Value of n is dependent on the retention rate percentage.

Summary generated by each algorithm for both datasets on different retention rate i.e., 30%, 40% and 50% has been evaluated individually and their scores were stored. These individual scores were further compared with the results of the survey paper[2].

We have done comparative analysis of the results obtained from both datasets for all the three algorithms. Also we have compared the results obtained of the CNN dataset with our base paper[2].

3.2 Architectural View

The proposed approach firstly retrieves articles from both of our dataset and then preprocess them by performing tokenization and removing the unwanted words. For Keyword extraction the three algorithm processes the document in their own way. Finally, according to retention rate top n sentences are selected for generating the summary. Figure 3.1 shows the overview of the system proposed in this project.

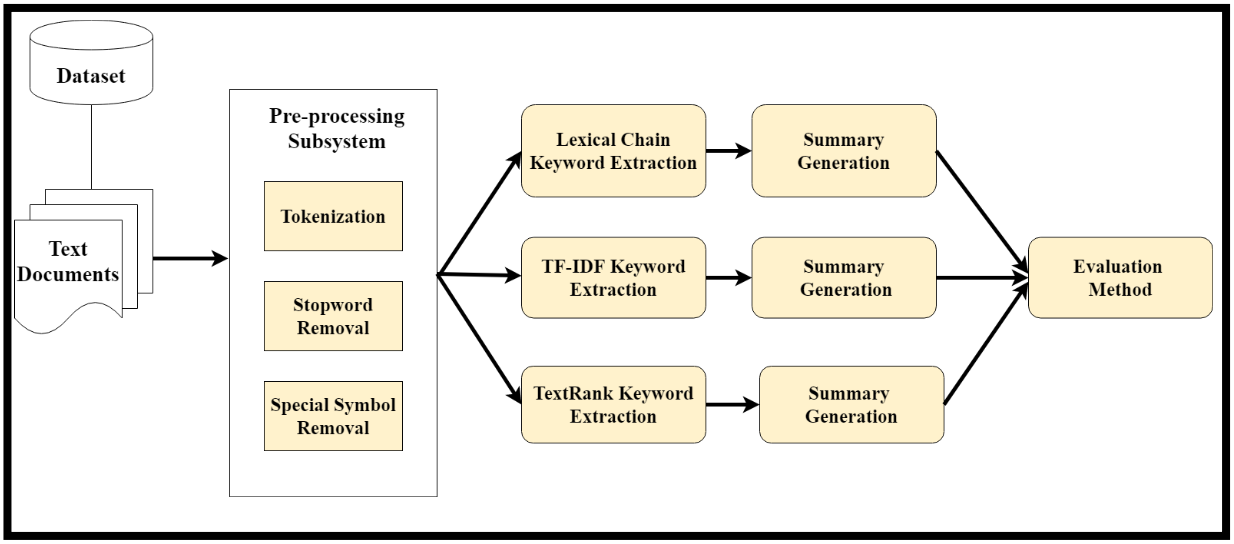

Figure 3.1 : Architectural View

3.3 Evaluation Method

ROUGE was used to quantitatively evaluate the summaries generated by using the different scoring methods. ROUGE is widely used for such purpose. This fully automated evaluator essentially measures the content similarity between system-developed summaries and the corresponding gold summaries.

The following five evaluation metrics are available.

- ROUGE-N: N-gram based co-occurrence statistics.
- ROUGE-L: Longest Common Subsequence based statistics. Longest common subsequence problem takes into account sentence level structure similarity naturally and identifies longest co-occurring in sequence n-grams automatically.
- ROUGE-W: Weighted LCS-based statistics that favors consecutive LCSes.
- ROUGE-S: Skip-bigram based co-occurrence statistics. Skip-bigram is any pair of words in their sentence order.
- ROUGE-SU: Skip-bigram plus unigram-based co-occurrence statistics.

For our experiments, we use the ROUGE-N metric, with N=1, since we are only concerned with single words extracted. Formally, ROUGE-N is an n-gram recall between a candidate summary and a set of reference summaries[11].

The general formula for calculating the metric is:

$$ROUGE-N = \frac{\sum_{S \in ref} \sum_{gram_n \in S} count_{match}(gram_n)}{\sum_{S \in ref} \sum_{gram_n \in S} count(gram_n)}$$

Where n stands for the length of the n-gram, $gram_n$, and $Count_{match}(gram_n)$ is the maximum number of n-grams co-occurring in a candidate summary and a set of reference summaries[11]. It is clear that ROUGE-N is a recall-related measure because the denominator of the equation is the total sum of the number of n-grams occurring at the reference summary side.

3.4 Chapter Summary

This chapter explains the architectural view of the proposed method.

Chapter 4 : Implementation

In this chapter we will discuss the experimental setup of the project done. First section will discuss the datasets. In the next section, implementation procedure is discussed followed by tools used for programming. In the last section summary of the chapter is given.

4.1 Dataset

To begin the project, there are certain preparatory steps which were taken for implementing the proposed model. First and foremost, was data collection. We required a moderately large amount of meaningful textual content since in our project we were assessing the performance of three algorithms on news paper articles. The two datasets that we selected are the CNN news dataset and BBC news dataset. Both the datasets were stored on Google Drive.

4.1.1 BBC News Dataset

We have used a public dataset of news articles from the BBC from 2004 to 2005 comprised of 2225 articles, each labeled under one of 5 categories: business, entertainment, politics, sport or tech. For each articles, golden summary is provided in the Summaries folder.

4.1.2 CNN News Dataset

The CNN corpus developed by Lins and colleagues (Lins et al., 2012) encompasses news articles from all over the world. The current version of this corpus presents 2000 texts assigned to 11 categories: Africa, Asia, business, Europe, Latin America, Middle East, US, sports, tech, travel, and world news. The texts were selected from the news articles of CNN website (http://www.cnn.com). Besides the very high quality, conciseness, general interest, up-to-date subject, clarity, and linguistic correctness, one of the advantages of this new corpus is that a good-quality summary for each text, called the highlights, is also provided. The highlights are three or four sentences long and are of paramount importance for evaluation purposes, as they may be taken as a summary of reference, or gold standard[2].

The reasons for selection of these dataset were the variety of non-related articles present in the dataset and the hand-written summaries of these articles provided with the dataset.

4.2 Algorithms

All the three algorithms were implemented independently for three different retention rates and the summaries generated by these algorithms were written in an output file that too was stored on Google drive and these summary files along with the golden summaries were passed into ROUGE for evaluation purpose. Algorithms proposed in Chapter 3 were used for generating the summaries and these algorithms are briefly explained in this section.

4.2.1 Lexical Chain

Given below is the flow chart for Lexical Chain.

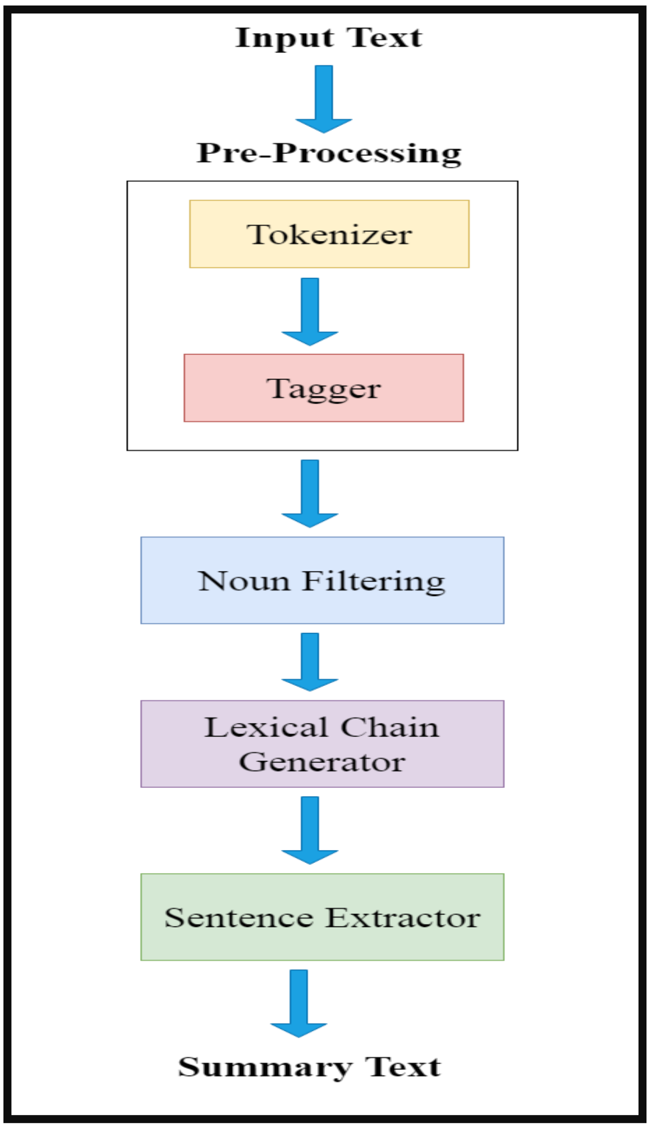

Figure 4.1 : Lexical Chain Flow Chart

Given below is the sample input news article.

```
Maria Sharapova has basically no friends as tennis players on the WTA Tour. The Russian player has no problems in openly
speaking about it and in a recent interview she said: 'I don't really hide any feelings too much. I think everyone knows
this is my job here. When I'm on the courts or when I'm on the court playing, I'm a competitor and I want to beat every
single person whether they're in the locker room or across the net.So I'm not the one to strike up a conversation about the
weather and know that in the next few minutes I have to go and try to win a tennis match. I'm a pretty competitive girl. I
say my hellos, but I'm not sending any players flowers as well. Uhm, I'm not really friendly or close to many players. I
have not a lot of friends away from the courts.' When she said she is not really close to a lot of players, is that
something strategic that she is doing? Is it different on the men's tour than the women's tour? 'No, not at all. I think
just because you're in the same sport doesn't mean that you have to be friends with everyone just because you're
categorized, you're a tennis player, so you're going to get along with tennis players. I think every person has different
interests. I have friends that have completely different jobs and interests, and I've met them in very different parts of
my life. I think everyone just thinks because we're tennis players we should be the greatest of friends. But ultimately
tennis is just a very small part of what we do. There are so many other things that we're interested in, that we do.'
```

Figure 4.2 : News Article

First, we performed the tokenization. Tokenization is the process of splitting a string, text into a list of tokens. One can think of token as parts, like a word is a token in a sentence, and a sentence is a token in a paragraph.

Sentence Tokenization – Splitting sentences in the paragraph.

```
['Maria Sharapova has basically no friends as tennis players on the WTA Tour.', "The Russian player has no problems in openly speaking about it and in a recent
```

Figure 4.3 : Tokenization of sentences

Word Tokenization – Splitting words in a sentence.

```
[['Maria', 'Sharapova', 'has', 'basically', 'no', 'friends', 'as', 'tennis', 'players', 'on', 'the', 'WTA', 'Tour'], ['The', 'Russian', 'player', 'has', 'no',
```

Figure 4.4 : Tokenization of words

After tokenization, we did tagging. It is the process of marking up a word in a corpus to a corresponding part of a speech tag, based on its context and definition.

```
[[('Maria', 'NNP'), ('Sharapova', 'NNP'), ('has', 'VBZ'), ('basically', 'RB'), ('no', 'DT'), ('friends', 'NNS'), ('as', 'IN'), ('tennis', 'NN'), ('players', 'NN
```

Figure 4.5 : Tagging

After performing preprocessing, we created lexical chains based on the lexical relationships of synonymy, antonymy, and one level of hypernyms/hyponymy. First of all, we had to go through the text taking - only the words that were in Noun's position. For this we take all the tags that wordnet gives us : NN, NNS, NNP, NNPS.

```
['maria', 'sharapova', 'friends', 'tennis', 'players', 'wta', 'tour', 'russian', 'player', 'problems', 'interview', 'feelings', 'everyone', 'job', 'courts', 'co
```

Figure 4.6 : Noun Filtering

Once we had all the names we create lists of relationships. In these lists we have put all the synonyms, antonyms, hyponyms, hypernym of each of the words. To perform the lexical chain we have to compared three things for each word:

- All the nouns we have already in the chain. If there is a noun in the chain, we will have to add one more to the final count.

- If in the relations list of this noun, we find one word in the actual chain. We will append our noun in the same chain. Moreover, we compute a similarity for the two words. We have a threshold for determine if the two words can be in the same chain or not.

- If the nouns that we have already defined in each chain, we look in their relations and if the current noun is in that relation list we will append the noun to the same chain. Like in the other case, we compute a similarity between the two words.

```
Chain 1 : {'interests': 2}
Chain 2 : {'lot': 2}
Chain 3 : {'locker': 1, 'room': 1}
Chain 4 : {'courts': 2, 'court': 1}
Chain 5 : {'job': 1, 'jobs': 1}
Chain 6 : {'everyone': 3}
Chain 7 : {'tour': 3}
Chain 8 : {'players': 1, 'player': 1}
Chain 9 : {'tennis': 5}
Chain 10 : {'friends': 5, 'person': 1, 'match': 1, 'girl': 1, 'players': 1, 'women': 1, 'player': 1}
```

Figure 4.7 : Lexical Chains

To make the summary we used lexical chain. We have computed the frequency of all the words in the text, taking into account that if a word is in a chain we count the sum of the whole chain. Once all the frequencies have been computed, we have normalized the values and performed filtering with a threshold both for maximum and minimum.

As a final result we had put in order the n most important sentences of our text that contained the largest amount of lexical chain and most frequent words in our text. In our case, value of n is based on the retention rate percentage that we have taken i.e., we have picked the top n most important sentences to generate the output summary.

Number of sentences were calculated as follows:

n= ((a * b) / 100)

a: percentage of retention rate i.e., 30,40 or 50

b: total no. of sentences present in the article

n: total no. of sentences in the final summary

```
Percentage of information to retain(in percent):40
No of sentences in final summary :  6
```

Figure 4.8 : No. of sentences in final summary

Given below is the generated summary for the mentioned sample input article on 40-percent retention rate using lexical chains.

```
When I'm on the courts or when I'm on the court playing, I'm a competitor and I want to beat every single person whether
they're in the locker room or across the net.So I'm not the one to strike up a conversation about the weather and know that
in the next few minutes I have to go and try to win a tennis match.
Uhm, I'm not really friendly or close to many players.
Is it different on the men's tour than the women's tour?
I think just because you're in the same sport doesn't mean that you have to be friends with everyone just because you're
categorized, you're a tennis player, so you're going to get along with tennis players.
I think everyone just thinks because we're tennis players we should be the greatest of friends.
```

Figure 4.9 : Generated Summary

4.2.2 TF-IDF

Given below is the flow chart for TF-IDF.

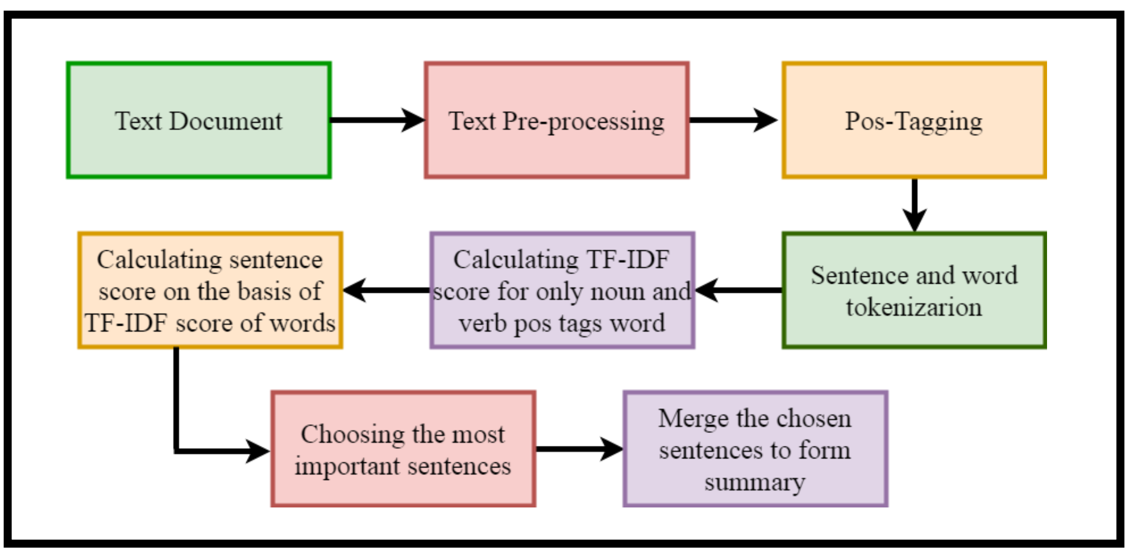

Figure 4.10 : TF-IDF Flow Chart

Below is the input article.

```
Maria Sharapova has basically no friends as tennis players on the WTA Tour. The Russian player has no problems in openly
speaking about it and in a recent interview she said: 'I don't really hide any feelings too much. I think everyone knows
this is my job here. When I'm on the courts or when I'm on the court playing, I'm a competitor and I want to beat every
single person whether they're in the locker room or across the net.So I'm not the one to strike up a conversation about the
weather and know that in the next few minutes I have to go and try to win a tennis match. I'm a pretty competitive girl. I
say my hellos, but I'm not sending any players flowers as well. Uhm, I'm not really friendly or close to many players. I
have not a lot of friends away from the courts.' When she said she is not really close to a lot of players, is that
something strategic that she is doing? Is it different on the men's tour than the women's tour? 'No, not at all. I think
just because you're in the same sport doesn't mean that you have to be friends with everyone just because you're
categorized, you're a tennis player, so you're going to get along with tennis players. I think every person has different
interests. I have friends that have completely different jobs and interests, and I've met them in very different parts of
my life. I think everyone just thinks because we're tennis players we should be the greatest of friends. But ultimately
tennis is just a very small part of what we do. There are so many other things that we're interested in, that we do.'
```

Figure 4.11 : News Article

The first step includes, reading text from a file and perform sentence tokenization.

Sentence Tokenization is the process of identifying where the sentences start and end. It is used to treat individual sentences as separate entities and makes further processing of the text relatively easy.

```
['Maria Sharapova has basically no friends as tennis players on the WTA Tour.', "The Russian player has no problems in openly speaking about it and in a recent
```

Figure 4.12 : Tokenization of sentences

After sentence tokenization, special characters are removed from the text. It is important to remove digits from the document, which can be done using regular expression.

After eliminating special character and digits, the individual words can be tokenized. *Word tokenizer* tokenizes words into separate entities within a sentence. This step is especially important if you want to calculate the feature scores of individual words of a sentence for deducing important sentences in a document.

```
[['Maria', 'Sharapova', 'has', 'basically', 'no', 'friends', 'as', 'tennis', 'players', 'on', 'the', 'WTA', 'Tour'], ['The', 'Russian', 'player', 'has', 'no',
```

Figure 4.13 : Tokenization of words

To avoid any ambiguity in case, we lowercase all the tokenized words and one letter word, stop words can be removed.

```
['maria', 'sharapova', 'basically', 'friend', 'tennis', 'player', 'wta', 'tour', 'the', 'russian', 'player', 'problem', 'openly', 'speaking', 'recent', 'intervi
```

Figure 4.14 : Conversion to lowercase

While working with text it becomes important to calculate the frequency of words, to find the most common or least common words based on the requirement of the algorithm.


```
{'maria': 1, 'sharapova': 1, 'basically': 1, 'friend': 5, 'tennis': 6, 'player': 8, 'wta': 1, 'tour': 3, 'the': 1, 'russian': 1, 'problem': 1, 'openly': 1, 'spe
```


Figure 4.15 : Calculate word frequency

Tagging of words returns only the noun and verb tagged words. The returned words are sent to word_tfidf function to calculate the score of that word in the document by calculating its TF-IDF score.

TF-IDF score is calculated by simply multiplying the TF and IDF values.

TF score is calculated as the number of times the word appears in the sentence upon the total number of words in the sentence.

IDF score of the word, by dividing the total number of sentences by number of sentences containing the word and then taking a $\log_{10}$ of that value.

```
TF Score, IDF Score, TF-IDF Score :
0.003289473684210526
1.2304489213782739
0.004047529346639059

Sentence Score :
0.02940954402642845

TF Score, IDF Score, TF-IDF Score :
0.003289473684210526
1.2304489213782739
0.004047529346639059

Sentence Score :
0.03345707337306751

TF Score, IDF Score, TF-IDF Score :
0.003289473684210526
1.2304489213782739
0.004047529346639059

Sentence Score :
0.03750460271970657

TF Score, IDF Score, TF-IDF Score :
0.0
1.2304489213782739
0.0
```

Figure 4.16 : TF, IDF, TF-IDF Scores

To find the most important sentences, take the individual sentences from tokenized sentences and compute the sentence score. After calculating the scores, the top n sentences based on the retention rate selected in the summary.

Number of sentences were calculated as follows:

n= ((a * b) / 100)

a: percentage of retention rate i.e., 30,40 or 50

b: total no. of sentences present in the article

n: total no. of sentences in the final summary

```
Percentage of information to retain(in percent):40
No of sentences in final summary :  6
```

Figure 4.17 : No. of sentences in final summary

```
Sentence No., Sentence Score
[(12, 0.15752869496446234), (4, 0.07469042250188664), (10, 0.07014482458221077), (3, 0.055027036248171804), (2, 0.04519250356950294), (5, 0.04394460433493835)
```

Figure 4.18 : Top N Sentences with scores

Given below is the generated summary using TF-IDF for the mentioned sample input article on 40-percent retention rate using TF-IDF.

```
The Russian player has no problems in openly speaking about it and in a
recent interview she said: 'I don't really hide any feelings too much.
I think everyone knows this is my job here.
When I'm on the courts or when I'm on the court playing, I'm a competitor
and I want to beat every single person whether they're in the locker room
or across the net.So I'm not the one to strike up a conversation about the
weather and know that in the next few minutes I have to go and try to win
a tennis match.
I'm a pretty competitive girl.
Is it different on the men's tour than the women's tour?
I think just because you're in the same sport doesn't mean that you have
to be friends with everyone just because you're categorized, you're a
tennis player, so you're going to get along with tennis players.
```

Figure 4.19 : Generated Summary

4.2.3 TextRank

Given below is the flow chart for TextRank. The first step would be to split the text into individual sentences. In the next step, we will find vector representation (word embeddings) for each and every sentence. Similarities between sentence vectors are then calculated and stored in a matrix. The similarity matrix is then converted into a graph, with sentences as vertices and similarity scores as edges, for sentence rank calculation. Finally, a certain number of top-ranked sentences n form the final summary[3].

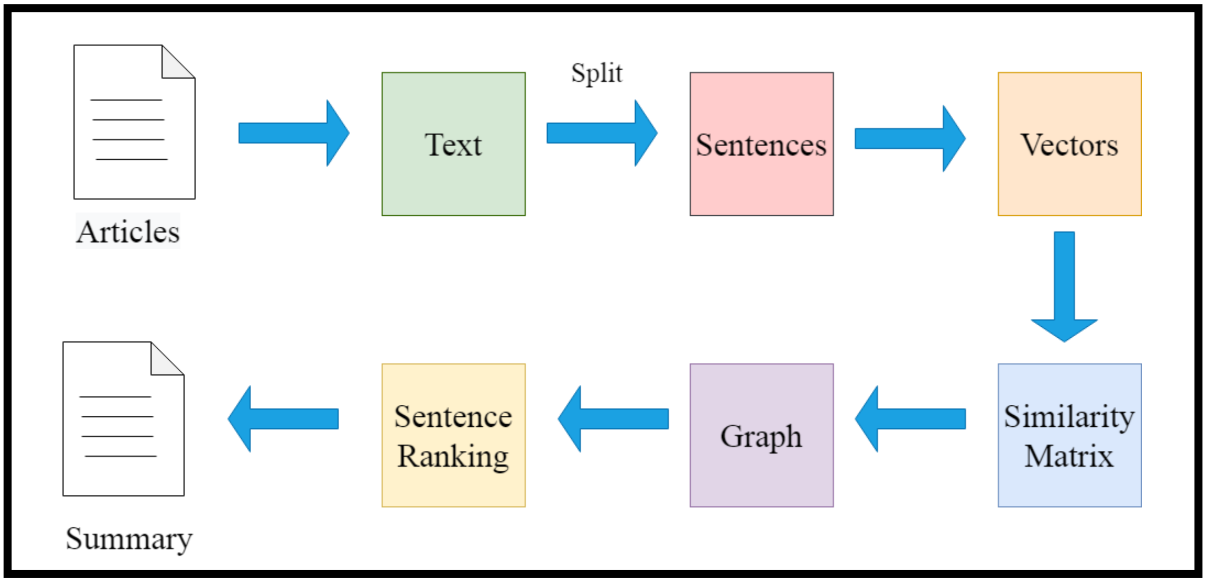

Figure 4.20 : TextRank Flow Chart

Given below is the sample input news article.

Maria Sharapova has basically no friends as tennis players on the WTA Tour. The Russian player has no problems in openly speaking about it and in a recent interview she said: 'I don't really hide any feelings too much. I think everyone knows this is my job here. When I'm on the courts or when I'm on the court playing, I'm a competitor and I want to beat every single person whether they're in the locker room or across the net.So I'm not the one to strike up a conversation about the weather and know that in the next few minutes I have to go and try to win a tennis match. I'm a pretty competitive girl. I say my hellos, but I'm not sending any players flowers as well. Uhm, I'm not really friendly or close to many players. I have not a lot of friends away from the courts.' When she said she is not really close to a lot of players, is that something strategic that she is doing? Is it different on the men's tour than the women's tour? 'No, not at all. I think just because you're in the same sport doesn't mean that you have to be friends with everyone just because you're categorized, you're a tennis player, so you're going to get along with tennis players. I think every person has different interests. I have friends that have completely different jobs and interests, and I've met them in very different parts of my life. I think everyone just thinks because we're tennis players we should be the greatest of friends. But ultimately tennis is just a very small part of what we do. There are so many other things that we're interested in, that we do.'

Figure 4.21 : News Article

Download GloVe Word Embeddings: GloVe word embeddings are vector representation of words. These word embeddings will be used to create vectors for our sentences. We could have also used the Bag-of-Words or TF-IDF approaches to create features for our sentences, but these methods ignore the order of the words (and the number of features is usually pretty large). We have used the pre-trained Wikipedia 2014 + Gigaword 5 GloVe vectors. The size of these word embeddings is 822

MB. We extract the words embeddings or word vectors and then we have word vectors for 400,000 different terms stored in the dictionary.

```
[1] !wget http://nlp.stanford.edu/data/glove.6B.zip
    !unzip glove*.zip

--2020-06-16 10:59:40--  http://nlp.stanford.edu/data/glove.6B.zip
Resolving nlp.stanford.edu (nlp.stanford.edu)... 171.64.67.140
Connecting to nlp.stanford.edu (nlp.stanford.edu)|171.64.67.140|:80... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://nlp.stanford.edu/data/glove.6B.zip [following]
--2020-06-16 10:59:41--  https://nlp.stanford.edu/data/glove.6B.zip
Connecting to nlp.stanford.edu (nlp.stanford.edu)|171.64.67.140|:443... connected.
HTTP request sent, awaiting response... 301 Moved Permanently
Location: http://downloads.cs.stanford.edu/nlp/data/glove.6B.zip [following]
--2020-06-16 10:59:41--  http://downloads.cs.stanford.edu/nlp/data/glove.6B.zip
Resolving downloads.cs.stanford.edu (downloads.cs.stanford.edu)... 171.64.64.22
Connecting to downloads.cs.stanford.edu (downloads.cs.stanford.edu)|171.64.64.22|:80... connected.
HTTP request sent, awaiting response... 200 OK
Length: 862182613 (822M) [application/zip]
Saving to: 'glove.6B.zip'

glove.6B.zip        100%[===================>] 822.24M  2.16MB/s    in 6m 29s

2020-06-16 11:06:11 (2.11 MB/s) - 'glove.6B.zip' saved [862182613/862182613]
```

Figure 4.22 : Download GloVe Word Embeddings

```
Length of word embedding :
400000
```

Figure 4.23 : Total Word Vectors in the Dictionary

Split Text into Sentences: Now the next step is to break the text into individual sentences. We have used the sent_tokenize( ) function of the nltk library to do this.

```
['Maria Sharapova has basically no friends as tennis players on the WTA Tour.', "The Russian player has no problems in openly speaking about it and in a recent
```

Figure 4.24 : Tokenization of Sentences

Text Preprocessing: It is always a good practice to make your textual data noise-free as much as possible. So, we have done some basic text cleaning. We removed punctuations, numbers and special characters and also made alphabets lowercase. We also removed stop words (commonly used words of a language – is, am, the, of, in, etc.) present in the sentences.

```
['maria sharapova basically friends tennis players wta tour', 'russian player problems openly speaking recent interview said really hide feelings much', 'think
```

Figure 4.25 : Clean Sentences (Sentences after Pre-processing)

Vector Representation of Sentences: We have used clean sentences(sentences come after preprocessing) to create vectors for sentences in our data with the help of the GloVe word vectors. To create vectors for our sentences, we first fetched vectors (each of size 100 elements) for the constituent words in a sentence and then take mean/average of those vectors to arrive at a consolidated vector for the sentence.

```
Sentence Vector :
[array([ 5.14825583e-02,  1.10544682e-01,  6.94999397e-01,  1.89168096e-01,
       -9.58077684e-02,  3.20288986e-01,  2.70662010e-01,  5.42440832e-01,
       -3.05938005e-01, -1.56364068e-01,  3.70127618e-01,  8.09492469e-02,
        8.41393881e-03,  2.47571543e-01, -3.69342804e-01, -7.61044994e-02,
        8.08582604e-02,  2.30643645e-01, -2.70402402e-01,  5.13828397e-01,
       -6.12548441e-02,  3.87900352e-01,  1.03121363e-01,  7.72494674e-01,
        2.59960234e-01, -7.96069205e-02,  1.42143592e-01, -9.62644577e-01,
        7.54904330e-01,  6.03260659e-02, -4.58570123e-01,  2.36780301e-01,
        2.29152635e-01, -1.56453326e-01,  3.97632688e-01, -2.32720934e-02,
       -5.05520999e-01,  4.13252831e-01, -2.85759270e-01, -1.35231465e-01,
       -1.37098104e-01, -1.48972601e-01,  3.37537557e-01, -3.49540442e-01,
        1.53484434e-01, -2.33341649e-01, -1.98460802e-01, -1.27821520e-01,
        5.08063912e-01, -3.68636876e-01, -2.28472307e-01, -3.15306723e-01,
        1.36149466e-01,  2.22507194e-01,  1.19500056e-01, -1.71007359e+00,
       -1.04403630e-01,  3.45346779e-01,  5.54419458e-01,  7.91236103e-01,
       -2.63593912e-01,  5.01183808e-01, -1.54918820e-01,  2.39762396e-01,
       -4.94388118e-02, -1.39404953e-01, -6.96142530e-03,  4.52243447e-01,
        1.44498184e-01, -1.88242078e-01,  1.62694290e-01,  2.51032356e-02,
       -1.29925504e-01, -2.16523811e-01, -1.39851749e-01,  1.79708660e-01,
```

Figure 4.26 : Sentence Vector

Similarity Matrix Preparation: The next step is to find similarities between the sentences, and we have used the cosine similarity approach to find similarities among the sentences. We created an empty similarity matrix for this task and populate it with cosine similarities of the sentences. We first defined a zero matrix of dimensions n* n. We initialized this matrix with cosine similarity scores of the sentences. Here, **n** is the number of sentences in the original article.

```
Similiarity Matrix :
[[0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0.]]
```

Figure 4.27 : Empty Similarity Matrix

```
Similiarity Matrix :
[[0.         0.64378339 0.59156996 0.7475453  0.63747269 0.61883837
  0.69372612 0.68542236 0.64739478 0.79788285 0.         0.83178782
  0.55843717 0.58564365 0.79771841 0.7565254  0.575041  ]
 [0.64378339 0.         0.83267683 0.85698557 0.71764094 0.81240582
  0.8420344  0.81021047 0.90473139 0.71734643 0.         0.8244583
  0.85043436 0.85658038 0.82539409 0.78656042 0.84938735]
 [0.59156996 0.83267683 0.         0.82313865 0.75527066 0.73470169
  0.79021472 0.80083597 0.89334065 0.60797113 0.         0.84927762
  0.8517375  0.75710064 0.89685303 0.69299048 0.84113187]
 [0.7475453  0.85698557 0.82313865 0.         0.76459354 0.82157695
  0.86001402 0.87651867 0.88054091 0.78653705 0.         0.91478753
  0.86048663 0.82472861 0.86071944 0.85267258 0.77808994]
 [0.63747269 0.71764094 0.75527066 0.76459354 0.         0.68546808
  0.74824065 0.72055054 0.76297432 0.66111451 0.         0.80672944
  0.75617725 0.71717036 0.77864099 0.72995365 0.71190387]
 [0.61883837 0.81240582 0.73470169 0.82157695 0.68546808 0.
  0.84673887 0.81307095 0.84076226 0.70893741 0.         0.82324845
  0.77039582 0.80494732 0.7927357  0.74317771 0.82921433]
 [0.69372612 0.8420344  0.79021472 0.86001402 0.74824065 0.84673887
  0.         0.83118278 0.8988784  0.73612404 0.         0.88512784
  0.8125782  0.83526862 0.86516029 0.79789072 0.84964389]
 [0.68542236 0.81021047 0.80083597 0.87651867 0.72055054 0.81307095
  0.83118278 0.         0.84124988 0.71694529 0.         0.87815398
  0.82030129 0.85208088 0.85451305 0.81738484 0.84847867]
```

Figure 4.28 : Similarity Matrix having Cosine Similarity Scores

Applying PageRank Algorithm: We have converted the similarity matrix into a graph. The **nodes** of this graph represent the sentences and the **edges** represent the similarity scores between the sentences. On this graph, we have applied the PageRank algorithm to arrive at the sentence rankings.

```
Apply Page Rank :
{0: 0.05418868552905123, 1: 0.06291967979381004, 2: 0.06107074622671161, 3: 0.06462055979515273, 4: 0.057678951524171324, 5: 0.06075695308614737, 6: 0.06364813003
```

Figure 4.29 : Sentence Score after applying PageRank

```
Ranked sentence :
[(0.06606217237750724, "I think just because you're in the same sport doesn't mean that you have to be friends with everyone just because you're categorized, you
```

Figure 4.30 : Sentence Ranking

Summary Extraction: Finally, extracted the top n sentences based on their rankings for summary generation. Number of sentences n were calculated as follows:

n= ((a * b) / 100)

a: percentage of retention rate i.e., 30,40 or 50

b: total no. of sentences present in the article

n: total no. of sentences in the final summary

```
Percentage of information to retain(in percent):40
No of sentences in final summary :  6
```

Figure 4.31 : No. of Sentences in the final summary

Given below is the generated summary for the mentioned sample input article on 40-percent retention rate using TextRank.

```
When I'm on the courts or when I'm on the court playing, I'm a competitor
and I want to beat every single person whether they're in the locker room
or across the net.So I'm not the one to strike up a conversation about the
weather and know that in the next few minutes I have to go and try to win
a tennis match.Uhm, I'm not really friendly or close to many players.I
have not a lot of friends away from the courts.'When she said she is not
really close to a lot of players, is that something strategic that she is
doing?I think just because you're in the same sport doesn't mean that you
have to be friends with everyone just because you're categorized, you're a
tennis player, so you're going to get along with tennis players.I think
everyone just thinks because we're tennis players we should be the
greatest of friends.
```

Figure 4.32 : Generated Summary

4.3 Evaluation Framework

As stated earlier, ROUGE is essentially of a set of metrics and works by comparing an automatically produced summary or translation against a set of reference summaries (golden summaries, typically human produced). We have used ROUGE-1 to evaluate the results. Results are provided and compared in the next chapter.

<u>How ROUGE evaluates the summary is discussed here:</u>

Let us say, we have the following system and reference summaries:

- ***System Summary (what the machine produced)*** **:** "the cat was found under the bed".
- ***Reference Summary (gold standard – usually by humans)*** **:** "the cat was under the bed".

Consider just the individual words, the number of overlapping words between the system summary and reference summary is 6. This however, does not tell much as a metric. To get a good quantitative value, the precision and recall using the overlap are computed by ROUGE[18].

Precision and Recall in the Context of ROUGE:

Recall in the context of ROUGE means how much of the reference summary is the system summary recovering or capturing. If considering the individual words, it can be computed as:

$$\frac{number\ of\ overlapping\ words}{total\ words\ in\ reference\ summary}$$

In this example, the Recall would thus be:

$$Recall = \frac{6}{6} = 1.0$$

This means that all the words in the reference summary has been captured by the system summary, which indeed is the case for this example. It looks really good for a text summarization system. However, it does not tell you the other side of the story. A machine generated summary (system summary) can be extremely long, capturing all words in the reference summary. But, much of the words in the system summary may be useless, making the summary unnecessarily verbose. This is where precision comes into play. In terms of precision, what you are essentially measuring is, how much of the system summary was in fact relevant or needed. Precision is measured as:

$$\frac{number\ of\ overlapping\ words}{total\ words\ in\ system\ summary}$$

In this example, the Precision would thus be

$$Precision = \frac{6}{7} = 0.86$$

This simply means that 6 out of the 7 words in the system summary were in fact relevant or needed. The precision aspect becomes really crucial when one are trying to generate summaries that are concise in nature. Therefore, it is always best to compute both the Precision and Recall and then report the F-Measure. If summaries are in some way forced to be concise through some constraints, then one could consider using just the Recall, since precision is of less concern in this scenario[18].

We have used ROUGE-1 which refers to overlap of unigrams between the system summary and reference summary. For extractive summarization with reference summaries, ROUGE-1 alone may suffice especially if stemming and stop word removal both are applied[18].

We run a python implementation of the three algorithms on both the dataset and stored the summaries for different retention rates on Google Drive. These stored summaries for three different retention rate were passed into ROUGE for calculating the F-measure, Recall and Precision(for ROUGE-1). The ROUGE-1 scores (F-measure, Recall and Precision) for both the datasets on three different retention rates were stored in a tabular form in CSV files on Google Drive. There are total of 18 CSV files (since all the three algorithms were run on each datasets, and each algorithm was run with three different retention rates. Since we have two datasets, total files are 18). After this, we find the average values of F-measure, Recall and Precision for both the datasets.

4.4 Programming Tools

We have run all the three algorithm on Google Colab and stored results on Google Drive.

Operating System	:	Windows
Language used	:	Python
Library used	:	NLTK, numpy, os, re, glob, networkx, csv, scikit-learn, ElementTree, ROUGE, google.colab
Platform	:	Google Colab

4.5 Chapter Summary

This chapter explains how the proposed method has been implemented.

Chapter 5 : Result And Analysis

In Chapter 5, we discuss and analyze the result we obtained.

5.1 Output

This section presents some abbreviations to better understand the experimentation and the results of the evaluation of the performance of the algorithms on different retention rate for both datasets.

5.1.1 Abbreviations

In order to facilitate presentation of the results, Table 5.1 lists the abbreviations to the terms used in next section and Table 5.2 shows a set of abbreviations for the name of algorithms.

Table 5.1 : Abbreviations

Average_R	Recall Average
Average_P	Precision Average
Average_F	F-measure Average
Alg	Algorithm

Table 5.2 : Algorithms

alg01	Lexical Chain
alg02	TF-IDF
alg03	TextRank

5.1.2 Results Obtained

The algorithms were run multiple times for three different retention rates for better understanding of the results on both datasets. The following table summarizes the results obtained for all the three algorithms on all the three retention rates.

5.1.2.1 Alg01

- **<u>FOR BBC NEWS DATASET</u>**

Table 5.3 : Results of ROUGE-1 for Alg01 on BBC News Dataset

Retention Rate	Average_R	Average_P	Average_F
30-percent	0.566264171	0.726121675	0.632109481
40-percent	0.702712356	0.684220102	0.690116081
50-percent	0.813495185	0.633500454	0.709658028

❖ **FOR CNN NEWS DATASET**

Table 5.4 : Results of ROUGE-1 for Alg01 on CNN News Dataset

Retention Rate	**Average_R**	**Average_P**	**Average_F**
30-percent	0.743792296	0.255915075	0.368077856
40-percent	0.819177644	0.220509198	0.337571139
50-percent	0.874737901	0.194504644	0.310349895

5.1.2.2 Alg02

❖ **FOR BBC NEWS DATASET**

Table 5.5 : Results of ROUGE-1 for Alg02 on BBC News Dataset

Retention Rate	**Average_R**	**Average_P**	**Average_F**
30-percent	0.340935626	0.555778908	0.417639816
40-percent	0.46642997	0.549426398	0.500277044
50-percent	0.589274442	0.534190631	0.556802366

❖ **FOR CNN NEWS DATASET**

Table 5.6 Results of ROUGE-1 for Alg02 on CNN News Dataset

Retention Rate	**Average_R**	**Average_P**	**Average_F**
30-percent	0.50371409	0.247482797	0.318157784
40-percent	0.619625226	0.219511077	0.313117923
50-percent	0.715029764	0.196682784	0.299440695

5.1.2.3 Alg03

❖ **FOR BBC NEWS DATASET**

Table 5.7 : Results of ROUGE-1 for Alg03 on BBC News Dataset

Retention Rate	**Average_R**	**Average_P**	**Average_F**
30-percent	0.536975678	0.73736768	0.61665669
40-percent	0.666104655	0.684419306	0.671599124
50-percent	0.773510216	0.629581347	0.691392266

❖ **FOR CNN NEWS DATASET**

Table 5.8 : Results of ROUGE-1 for Alg03 on CNN News Dataset

Retention Rate	**Average_R**	**Average_P**	**Average_F**
30-percent	0.702114445	0.252875909	0.358490496
40-percent	0.783297145	0.218070958	0.330701526
50-percent	0.846266845	0.192953576	0.305842397

5.2 Assessment

The assessment was performed using each dataset separately. For both the datasets, the average values of F-measure, Recall and Precision were calculated on three retention rates per algorithm and shown in the Table 5.9 and Table 5.10.

5.2.1 Assessment Using BBC News Dataset

The average result of calculating ROUGE for each algorithm using BBC News Dataset is shown in Table 5.9. Although results obtained were closed, some points should be remarked:

- Alg01 achieved the best Recall.
- Alg03 achieved the best Precision.
- Alg01 also achieved the best F-measure.

Out of all three algorithm, **the best summarization results on BBC News Dataset is provided by Alg01.**

Table 5.9 : Average Results of algorithms on BBC News Dataset

Algorithm	**Average_R**	**Average_P**	**Average_F**
alg01	0.694157237	0.681280743	0.67729453
alg02	0.46554668	0.546465313	0.491573077
alg03	0.658863517	0.683789443	0.659882693

Accuracy of an algorithm has been calculated by taking the average of all the parameters : F-measure, Recall and Precision values. Following Figure 5.1 shows the graphical representation of these results on BBC News Dataset.

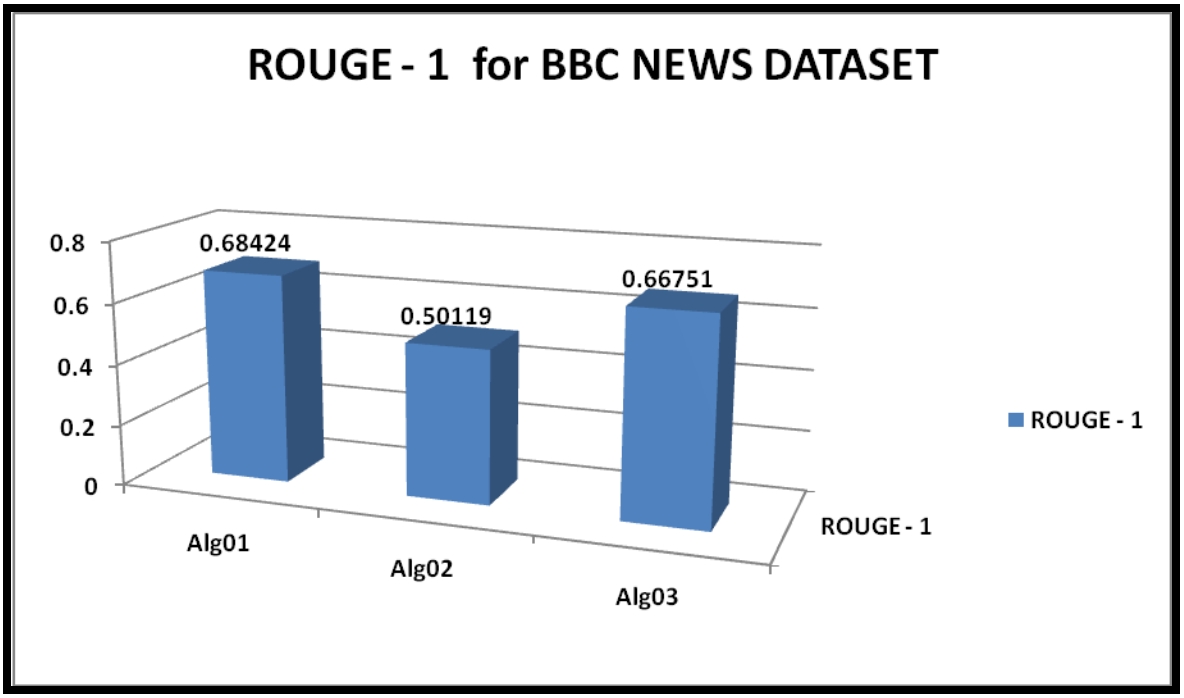

Figure 5.1 : Comparison of algorithms on BBC Dataset using ROUGE-1

5.2.2 Assessment Using CNN News Dataset

The average result of calculating ROUGE for each algorithm using CNN News Dataset is shown in Table 5.10. Although results obtained are close, some points should be remarked:

- Alg01 achieved the best Recall.
- Alg01 achieved the best Precision.
- Alg01 also achieved the best F-measure.

Out of all three algorithm, **the best summarization results on CNN News Dataset is provided by Alg01.**

Table 5.10 : Average Results of algorithms on CNN News Dataset

Algorithm	Average_R	Average_P	Average_F
alg01	0.81256928	0.223642972	0.338666297
alg02	0.612789693	0.221225553	0.310238801
alg03	0.777226147	0.221300148	0.33167814

Accuracy of an algorithm has been calculated by taking the average of all the parameters : F-measure, Recall and Precision values. Following Figure 5.2 shows the graphical representation of these results on CNN News Dataset.

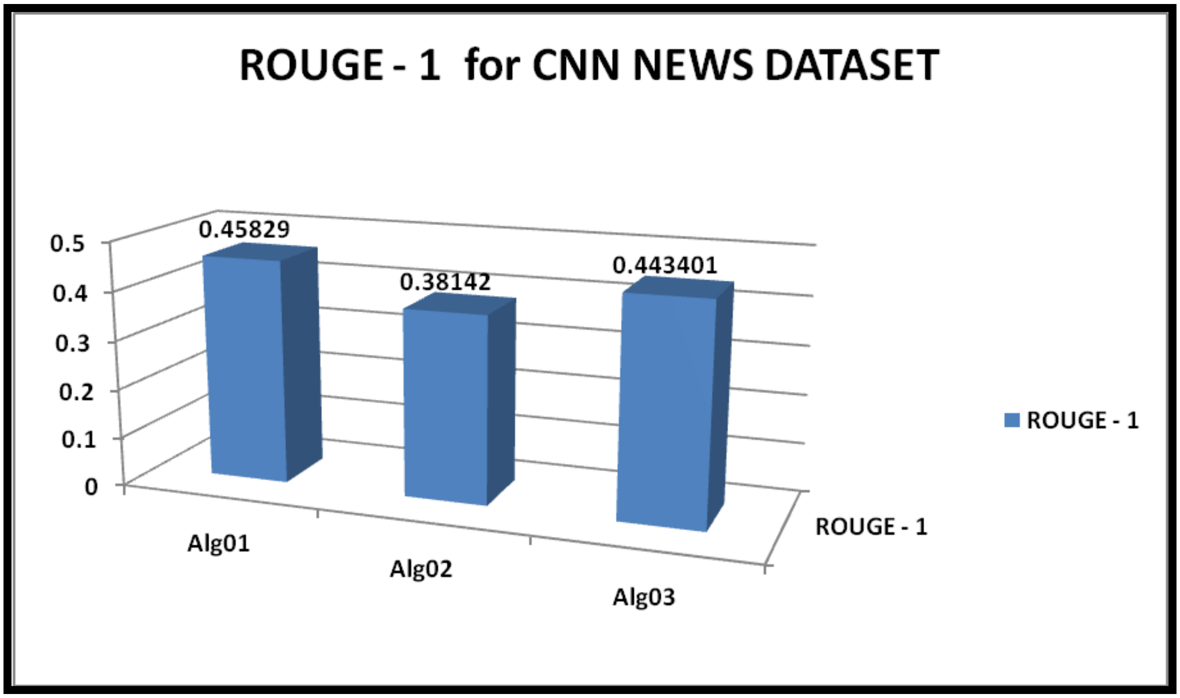

Figure 5.2 : Comparison of algorithms on CNN Dataset using ROUGE-1

5.3 Analysis

The quantitative assessment was done using ROUGE and then we performed the qualitative assessment over the results obtained. Retention rate tells us about the number of sentences in the final summary. More the retention rate, more will be the sentences in the summary.

In CNN News Dataset, the size of the articles were quite large. But in case of BBC News dataset, some of the articles were large while the other were very small in terms of sentences. Therefore, while generating the summaries for these small articles for retention rate at 10 and 20 percent the value were n was less than one, and hence the summary could not be generated. Therefore, we choose the retention rate from 30, 40 and 50 percent respectively.

We have found that the retention rate was directly proportional to Recall i.e., as we increased the retention rate the Recall for the particular algorithm also increased(as can be observed from table 3 to table 8). Also we have found that the retention rate was inversely proportional to the Precision i.e., as we increased the retention rate the Precision for a particular algorithm decreased(as we can observed from table 3 to table 8). Also we have found that we decreased the retention the readability and connectivity of the summary was somewhere lost. However while increasing the retention rate the number of lines in the summary increased which resulted in achieving a better summary both in terms of readability and connectivity each.

In survey paper[2], they have used only five to six sentences to generate the summary which were independent of the sentences present in the original article, whereas in our case we have generated the summary for different retention rates which give different values of n where n is the number of lines in the summary and therefore it is dependent on number of sentences in the original article which is one of the reason why our generated summaries were more readable and connected.

We observed that for all the three retention rates we were achieving quite good results, and among all the three retention rates we used, the summary generated with the retention rate 40 were best.

The results given in survey paper[2] for CNN News Dataset is shown in Table 5.11.

Table 5.11 : Results in Survey Paper for CNN News Dataset

Algorithm	Average_R	Average_P	Average_F
alg02	0.73	0.35	0.46
alg03	0.62	0.34	0.43

Chapter 6 : Conclusions And Future Work

6.1 Conclusions

This report gives the details of extractive text summarization for newspaper articles using three different algorithms i.e., Lexical Chain, TF-IDF, TextRank on both the benchmark datasets of BBC News Dataset and CNN News Dataset. We performed both quantitative and qualitative assessment on the results and also analyze the result.

6.2 Future Work

On the basis of our observation and analysis, we proposed following extension which can be incorporated to further improve the efficiency and accuracy of automatic summary of news articles: One can combine several sentence scoring algorithms into one single system which will incorporate the diverse features of these algorithms to produce better summary for news articles. This can explain as follows: we know that in news articles, starting lines and last lines of the article are very important as they contain most informative information, so we can give more score to these sentences, this could be done using sentence positioning algorithms. Also for news articles, we can give more scores to the sentence which resemblance with the title the most i.e., which has most keyword overlapping the title. For this, Resemblance with title algorithms can be used. Also Sentence inclusion of Numeric Data Algorithms can be combined with the above two algorithm to verify using regular expressions if some numeric data is present and give higher score to sentences in which numerical data is present. By combining the results of above three algorithm using some measure, a better summary exclusively for the news articles can be generated, since these algorithms covers most of the features of the newspaper article.

Also as a part of future work, one can further extend this work as follows:

- Multi document summarization - an automatic procedure aimed at extraction of information from multiple texts written about the same topic. This will give multiple dimensions to the user on single issue.
- News Summarization using deep learning methods - one can also generate the summaries using Recurrent Neural Network (RNN) and Long Short Term Techniques (LSTM).
- Cross-language summarization - means the original document is in some language **X** while the generated summary is in language **Y**. This could be useful for user who are interested in

different literature but learning these many literature is nearly impossible. Therefore, these system will produced the summaries of several literature in the language in which user is proficient.

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